

Components of Computers

Week 2/16

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Outline of Topics

1. What is hardware and software?
2. About the hardware
 - Central Processing Unit,
 - Arithmetic Logic Unit, Control Unit,
 - Input /Output Devices
3. Advantages and Disadvantages of Computers
4. CPU Power vs Memory Performance in terms of Instructions per Second
5. Number of Transistors per Microprocessor over time
6. Processor Clockspeed
7. Computing Efficiency (Wattage Consumed vs Instructions per Second)

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Learning Objectives:

At the end of the class, you should be able to:

- Define and distinguish between hardware and software
- Describe three fundamental categories of software and their relationship
- Discuss the factors that make a computer application a useful tool
- Describe the role of the operating system in a modern computer system
- Enumerate some of the capabilities and limitations of computers
- Write and submit your first program “Hello World...” using the [programming tools in 1940s ;\)](#)

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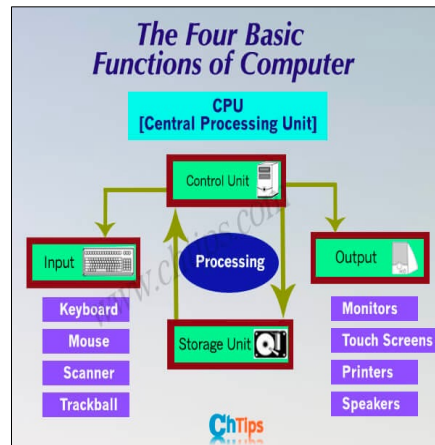
Definition of Modern Computer

- A computer is a general-purpose device that can be programmed to carry out a set of arithmetic or logical operations automatically.
- A computer is a programmable electronic device that processes, stores, and manipulates data to perform a wide range of tasks and calculations.
- Computers are used in various fields and applications, including business, communication, education, entertainment, research, and more.

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Basic Functions of a Computer

- **Receive input** -- accept information from the outside world (input devices such as the keyboard)
- **Store information** -- move and store information in memory (storage devices)
- **Process information** -- perform arithmetic or logical operations on information (CPU)
- **Produce output** -- communicate information to the outside world (output devices e.g. monitor)

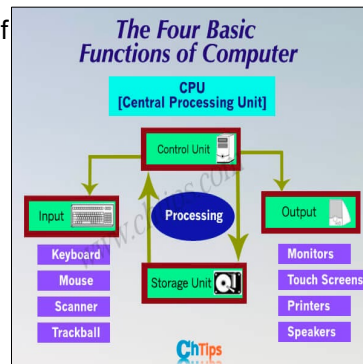


Source: <https://www.chtips.com/computer-fundamentals/functions-of-computer-system>

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Two major Components of a Computer System

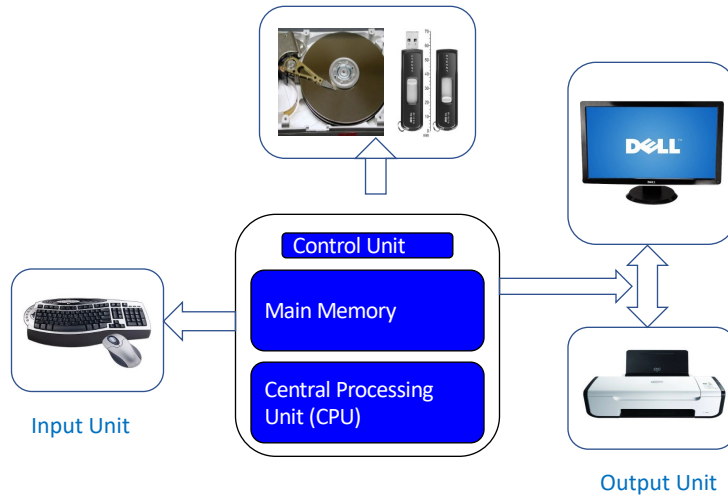
1. Hardware – is the physical parts of the of computer
 - a. Input devices – keyboard, mouse
 - b. Memory
 - Internal Memory – RAM & ROM
 - External Memory – hard disks, USB,
 - c. Central Processing Unit
 - Arithmetic Logic Unit
 - Control Unit
 - d. Output devices – monitor, printer, speakers
 - e. Peripherals
2. Software – the instruction and data of the computer



Source: <https://www.chtips.com/computer-fundamentals/functions-of-computer-system>

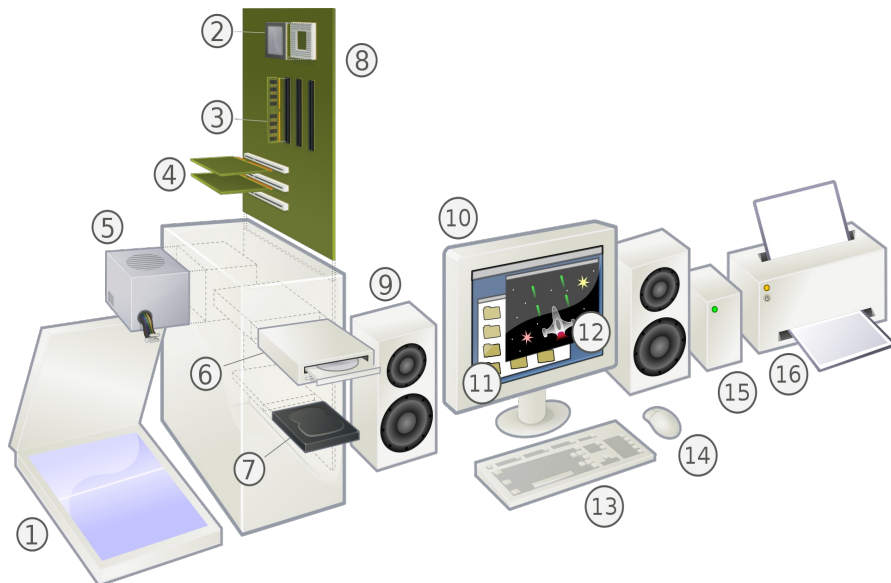
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The Computer Hardware



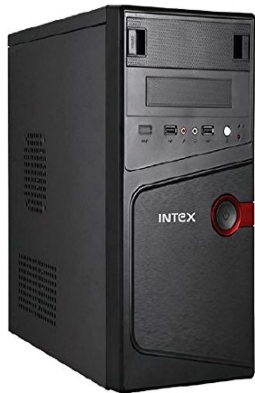
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The Computer Hardware



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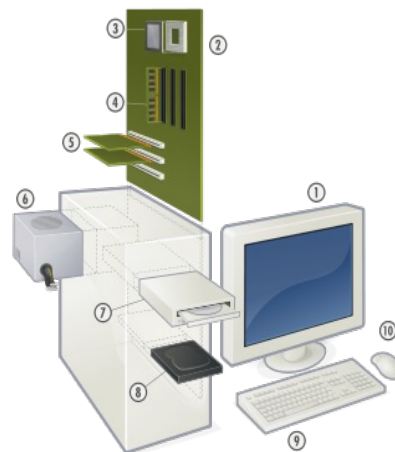
The Computer Hardware: CPU Box



Housed the CPU along with other chips and electronic components on the motherboard, power supply, hard drive, cd/dvd player, etc

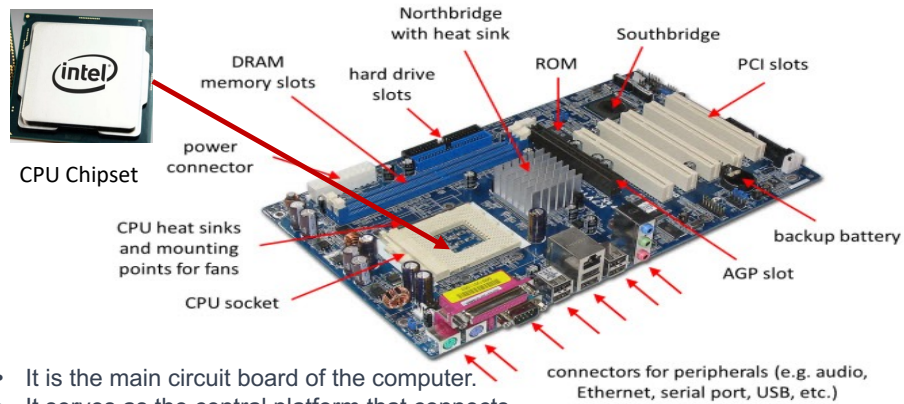
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The CPU Box



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The Main Motherboard

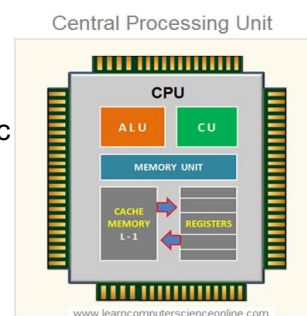


- It is the main circuit board of the computer.
- It serves as the central platform that connects and houses various hardware components, including the CPU, RAM, storage devices, and expansion cards.
- It provides power and data connections between components.

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Central Processing Unit

- The central processing unit (CPU) is the brain of the computer system
- It is a digital circuit used to perform arithmetic and logic operations.
- It is the core component of a computer responsible for executing instructions, performing calculations, and managing data.
- The CPU includes the following components:
 - Control Unit
 - ALU
 - Registers
 - Cache Memory



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Arithmetic Logic Unit

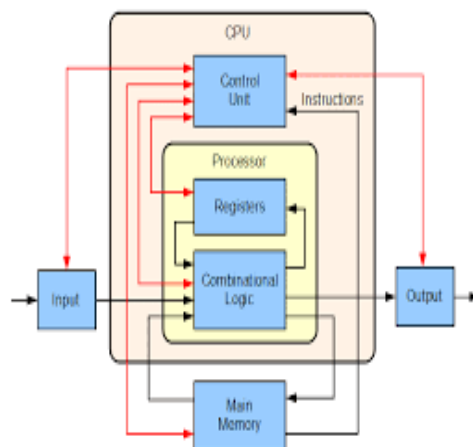
The main functions of the ALU:

1. Arithmetic Operations: +, -, /, *
2. Logical Operations: Logical AND, OR, NOT
3. Comparison Operations: ==, >=, >, <=, !=
4. Shift and Rotate Operations
5. Bitwise Operations
6. Overflow Detection
7. Flag Setting
8. Data Manipulation -- extracting specific fields, converting between data types, and performing various data transformations.

It works in conjunction with other CPU components such as registers and control units, to fetch, decode, execute, and store the results of instructions.

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The Control Unit



- It serves as the brain of the CPU, orchestrating the execution of instructions and data processing within the CPU.
- It ensures that instructions are executed in the correct order and according to the program's logic, playing a pivotal role in the computer's ability to process data and perform tasks efficiently.

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Functions of the Control Unit

1. Instruction Fetch -- retrieves instructions from memory based on the address provided by the Program Counter (PC), and initiates memory read operations to fetch the next instruction.
2. Instruction Decoding -- decodes the instruction to determine the operation to be performed, the operands involved, and the sequence of steps required for execution.
3. Instruction Execution -- oversees the execution of instructions by coordinating the appropriate arithmetic, logical, or data transfer operations within the CPU.
4. Control Signals Generation --- generates control signals that manage various components of the CPU, including the Arithmetic Logic Unit (ALU), registers, and data paths.
 - These signals dictate the specific operations to be performed and the flow of data within the CPU.
5. Error Detection and Handling -- monitors for errors during instruction execution, such as arithmetic

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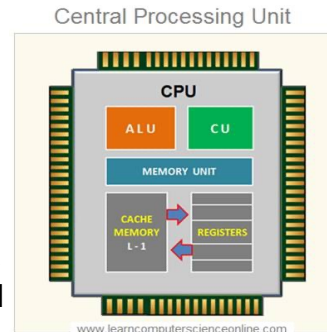
Functions of the Control Unit

6. Data Transfer -- manages the movement of data between components & controls the pathways & timing of data transfers to ensure data integrity.
7. Register Operations -- enables loading, storing, and manipulation of data in CPU registers
8. Branching and Control Flow -- handles conditional and unconditional branching instructions, making decisions based on comparison results and controlling the flow of program execution.
9. Clock Signal Management -- synchronizes CPU operations with the system clock, ensuring that each instruction and operation is executed in the correct sequence and within the appropriate clock cycle.
10. Interrupt Handling -- when external events or hardware devices generate interrupts, the CU suspends the current execution, saves the current context, and transfers control to an interrupt service routine.

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CPU's Cache Memory

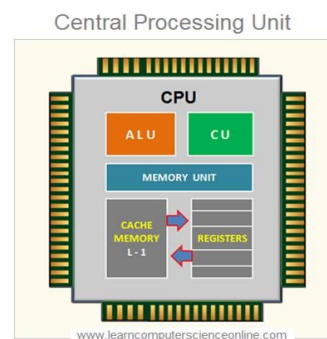
- CPU cache or processor cache
- is a high-speed, small-sized type of memory that is an integral part of the central processing unit (CPU) in modern computer architectures.
- Cache memory is designed to bridge the speed gap between the CPU and the main system memory (RAM) and serves as a temporary storage buffer for frequently accessed data and instructions.



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CPU's Cache Memory

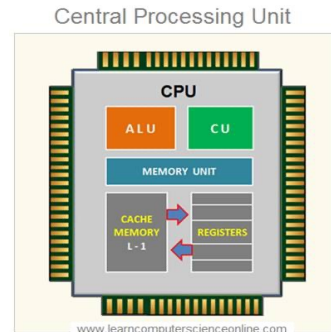
- It is distinct from RAM and serves a different purpose within the CPU.
- The primary purpose of cache memory is to store frequently used data and instructions that the CPU is likely to access in the near future.
- It acts as a buffer to provide faster access to this frequently used information, reducing the CPU's need to access data from slower RAM or storage devices.



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CPU's Registers

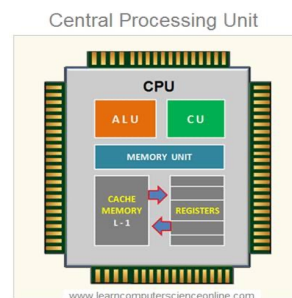
- The CPU contains several built-in registers, each serving a specific function and playing a crucial role in the execution of instructions and data processing.
- These are tiny, high-speed storage locations within the CPU, and they are used to temporarily hold data, addresses, and control information during various stages of the instruction execution process.



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CPU's Registers

- These CPU registers work together to facilitate the execution of instructions and the management of data and control flow within the CPU.
- They contribute to the speed and efficiency of the CPU's operations by providing quick access to the necessary data and control information.
- The specific registers and their functions can vary between CPU architectures and instruction set architectures.

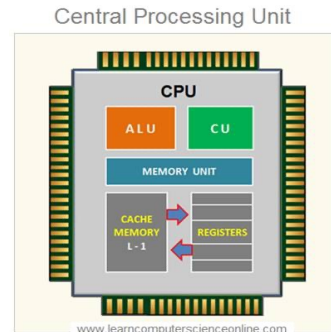


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CPU's Registers

• Type of Registers

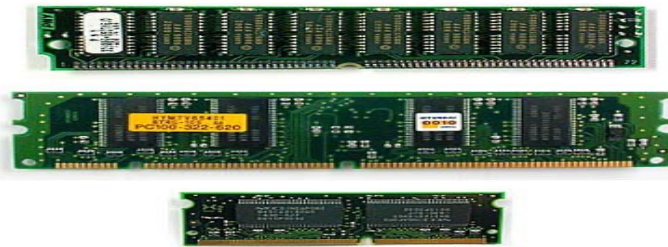
- Program Counter (PC)
- Instruction Register (IR)
- Memory Address Register (MAR)
- Memory Data Register (MDR)
- General-Purpose Registers
- Flag Registers (Status Registers)
- Stack Pointer (SP) and Base
- Index Registers
- Instruction Pointer (IP)



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Memory/ Storage

- Computer Memory – stores information of all types: instructions, numbers, names, lists and pictures.
- Types: RAM, ROM and Secondary Memory



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Random Access Memory (RAM)

- The RAM's primary function is to provide high-speed, temporary storage for data that the computer's CPU (Central Processing Unit) can quickly access and use.
- It is volatile/ temporary storage to temporarily store data that is actively being used or processed.
- When the computer is booted, the operating system is loaded into RAM.
- All programs, including the operating system, and any data are using the RAM for the computer to access them.
- When you open a program or file, it is loaded from storage (e.g., a hard drive or SSD) into RAM for fast processing.
- RAM size (the bigger, the better) and speed significantly impact a computer's performance.

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Random Access Memory (RAM)

Important of functions of RAM:



1. Temporary Data Storage
2. Program Execution
 - Program's code and data are loaded into RAM to allow the CPU to execute instructions.
 - This enables the program to run smoothly and respond quickly to user input.
3. Multitasking
 - RAM allows a computer to handle multiple tasks or programs simultaneously.
 - Each program or application running on the computer is allocated a portion of RAM, and the operating system manages the switching of data between RAM and the CPU to give the appearance of multitasking.
4. Operating System Functionality
 - OS uses RAM to store critical system data and processes such as managing system resources, running background tasks, and handling user interactions.

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Random Access Memory (RAM)

5. **Caching** -- used to store frequently accessed data and instructions, which can be quickly retrieved by the CPU.
5. **Virtual Memory** -- allows a computer to use a portion of the hard drive as if it were additional RAM.
 - When physical RAM becomes full, less frequently used data is swapped between RAM and the hard drive, ensuring that the system can continue to function efficiently even when it has more data to process than it can store in physical RAM alone.
7. **Data Transfer** -- acts as a temporary buffer for data being transferred between different hardware components, such as between the CPU and storage devices, or between the CPU and input/output devices like the graphics card.
8. **Quick Data Access** -- provides much faster data access times compared to storage devices like hard drives or SSDs, which is crucial for overall system performance, as it reduces delays when accessing and processing data.

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Read Only Memory (ROM)

- It is non-volatile, so contents are not lost when your computer is turned OFF.
- It stores data or instructions that are permanently written during manufacturing and cannot be easily modified by a user.
- Contains start-up instructions and other permanent data needed to start the computer
- It then begins loading the operating system found on the hard disk, in a special location, into RAM.



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Types of Read Only Memory (ROM)

1. BIOS (Basic Input/Output System)
 - a specific type of ROM in computers that contains a firmware
 - It is responsible for managing hardware initialization and the boot process/ loading the operating system.
 - It has the capability for updates to accommodate changes in hardware or system requirements.
2. Mask ROM (MROM)
 - A ROM that is programmed at the time of manufacturing, and its data is permanently etched onto the chip during production.
 - Its program cannot be altered or reprogrammed, and
 - it is often used for storing firmware or software that should remain unchanged.
3. EPROM (Erasable Programmable Read-Only Memory) -- a ROM that can be erased and reprogrammed using specialized equipment.
4. Flash Memory -- a type of EEPROM that can be electrically erased; commonly used for firmware updates and for storing data in devices like USB drives, SSDs, and memory cards.

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BIOS Instructions and Functions

1. Power-On Self-Test (POST)
 - performs a series of tests on the computer's hardware components, including the CPU, RAM, storage devices, graphics card, and more.
 - These tests ensure that the hardware is functioning correctly. If any issues are detected during POST, error codes or messages may be displayed to alert the user.
2. Initialization of Hardware
 - The BIOS initializes various hardware components, including setting up the CPU, configuring system memory (RAM), initializing the hard drive or SSD, initializing USB ports, and configuring other peripherals such as the keyboard and mouse.
3. Boot Device Selection
 - allows the user to specify the order in which the computer should attempt to boot from various storage devices (e.g., hard drive, CD/DVD drive, USB drive).
 - This is known as the boot order or boot sequence.

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BIOS Instructions and Functions

4. Loading the Boot Loader.

- Once the hardware is initialized, the BIOS locates and loads the boot loader.
- The boot loader is a small program stored on the bootable storage device (e.g., the Master Boot Record or EFI System Partition) that is responsible for loading the computer's operating system.

5. BIOS Setup Utility – BIOS configuration

- BIOS configuration allows the user to configure various system settings such as date and time, boot order, hardware settings, security options, and more.
- Access to the BIOS setup utility is typically achieved by pressing a specific key (e.g., Del, F2, F12) during the early stages of the boot process.

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BIOS Instructions and Functions

6. BIOS Password Security

- The BIOS can also support password protection to restrict access to the BIOS setup utility and prevent unauthorized changes to system settings.

7. CMOS (Complementary Metal-Oxide-Semiconductor) Memory

- The BIOS ROM communicates with a small amount of non-volatile memory known as CMOS memory.
- CMOS stores system configuration settings, including date and time, boot order, and hardware parameters.
- CMOS memory is powered by a battery to ensure it retains data even when the computer is turned off.



8. Error Handling and Reporting

- If any critical hardware errors occur during the boot process, the BIOS may display error messages or beep codes to help diagnose the issue.

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Secondary Memory/Storage

- Non-Volatile - makes it suitable for long-term storage.
- Long-Term Storage - used for storing data and programs that need to be preserved over extended periods; allows users to store files, documents, applications, and the operating system.
- High Capacity -- offer larger storage capacities compared to RAM
- Slower Access Speed - Accessing data from secondary memory is slower compared to accessing data from RAM.

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Secondary Memory/Storage



Common Secondary Memory/Storage

1. Hard Disk Drives (HDDs) -- provide high-capacity storage but are relatively slower than SSDs.
2. Solid-State Drives (SSDs) -- use NAND flash memory to store data and offer much faster read and write speeds compared to HDDs. They are also more durable and energy-efficient.
3. Optical Discs
 - Optical discs like CDs, DVDs, and Blu-ray discs are optical storage media
 - used for archiving data, movies, music, and software.
 - They are read-only (CD-ROM, DVD-ROM, BD-ROM) or rewritable (CD-RW, DVD-RW, BD-RE).
4. USB Flash Drives
 - These portable, flash-based storage devices offer a convenient way to store and transport data.
 - They are small, lightweight, and do not have moving parts.

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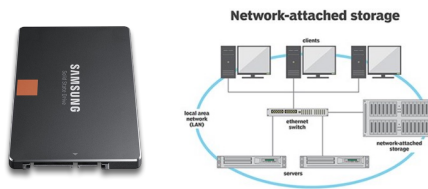
Secondary Memory/Storage

Common Secondary Memory/Storage

5. External Hard Drives

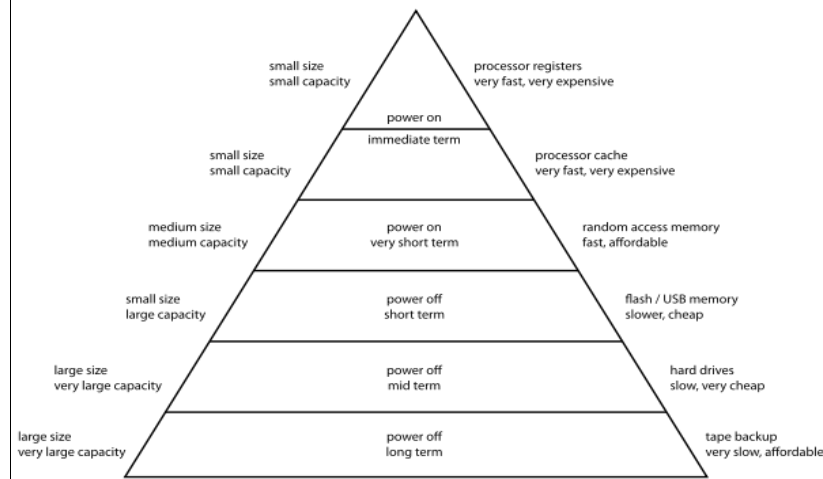
- External HDDs and SSDs are secondary storage devices that connect to a computer via USB, HDMI, Thunderbolt, or other interfaces.
- They are used for backup, data transfer, and additional storage capacity.

6. Network Attached Storage (NAS): NAS devices are specialized secondary storage solutions that connect to a network and provide shared storage for multiple users or devices. They are often used in home and office environments.



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Computer Memory Hierarchy



Source: https://en.wikipedia.org/wiki/Memory_hierarchy

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The I/O Devices

- Input and Output Devices – enables computer users to communicate with the computer

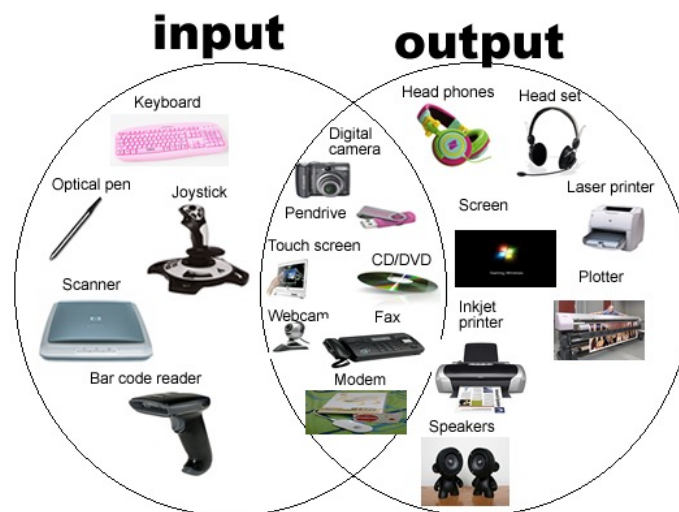
Input Devices



Output Devices

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The I/O Devices



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Buses, Ports, and Peripherals

- Information travels between components on the motherboard through groups of wires called **system buses**, or just **buses**

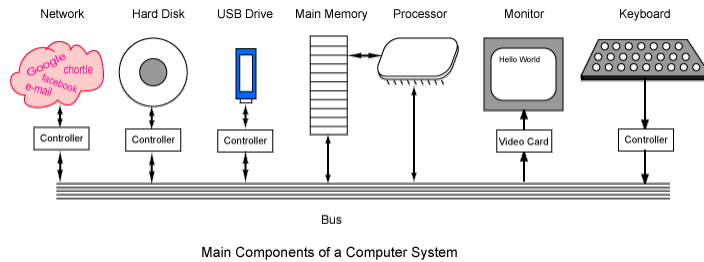


Image source: https://chortle.ccsu.edu/java5/Notes/chap01/ch01_4.html

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Buses, Ports, and Peripherals

Buses

- Typically have 32 or 64 wires
- Connect to storage devices in bays
- Connect to expansion slots
- Connect to external buses and ports

Slots and ports

- Make it easy to add external devices, called **peripherals**

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Advantages and Disadvantages of Computer

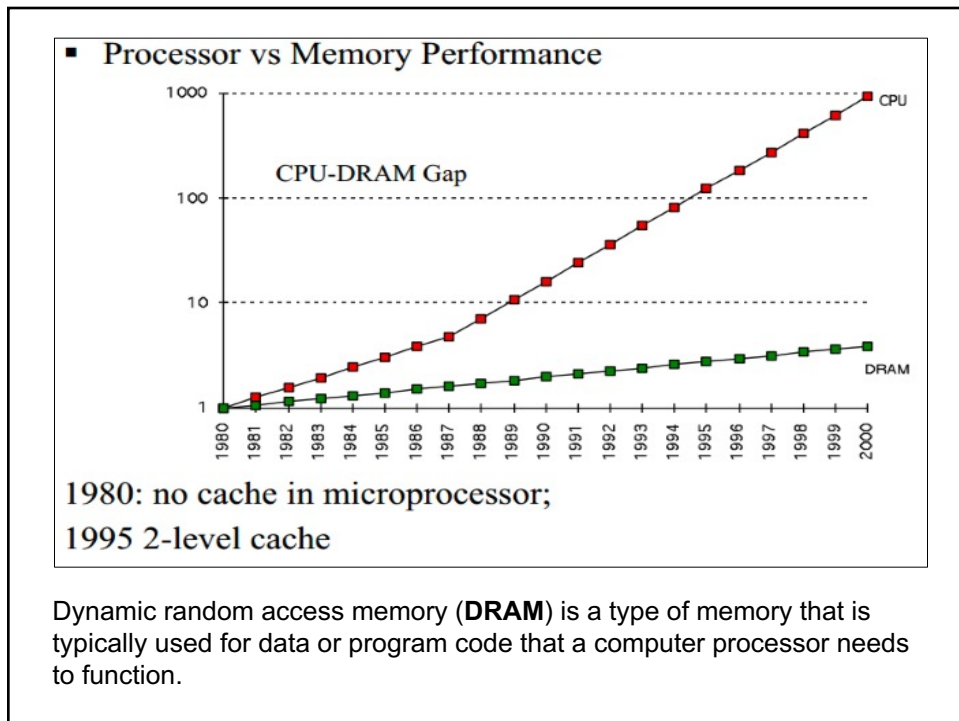
Advantages

- ✓ Speed
- ✓ Accuracy
- ✓ Stores Huge Amount of data
- ✓ Online Trading
- ✓ Online Education | Distance Learning
- ✓ Research
- ✓ Forecasting Weather, and Predicting
- ✓ Earthquakes, Volcano Eruptions
- ✓ Produce Employment
- ✓ Internet
- ✓ In Business

Disadvantages

- ✓ Health Issues.
- ✓ Spread of Pornography
- ✓ Virus and Hacking Attacks
- ✓ Computer can not take their own decision NO IQ
- ✓ Negative effect on Environment
- ✓ Crashed Networks
- ✓ Computer can not work on itself
- ✓ Spread of violence, hatred
- ✓ Online Cyber Crimes
- ✓ Data and Information violation

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Moore's Law: Transistors per microprocessor

Number of transistors which fit into a microprocessor. This relationship was famously related to Moore's Law, which was the observation that the number of transistors in a dense integrated circuit doubles approximately every two years.

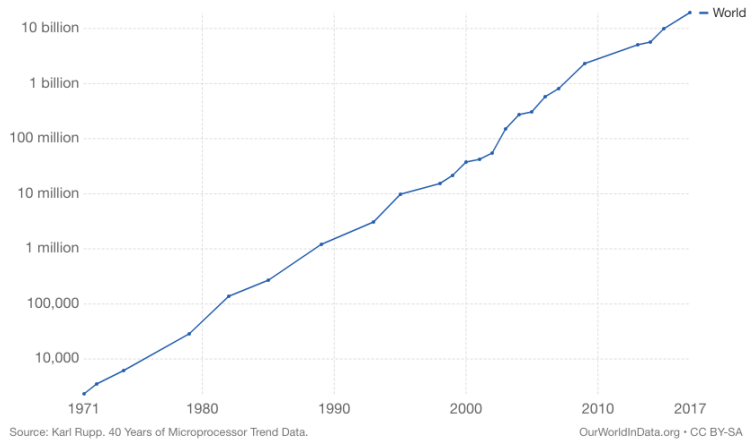


Image sources:

[https://en.wikipedia.org/wiki/Ignite_\(microprocessor\)](https://en.wikipedia.org/wiki/Ignite_(microprocessor))

<https://www.adafruit.com/product/976>

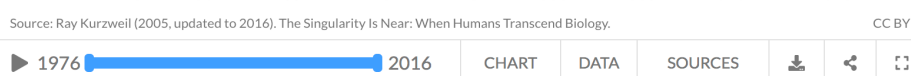
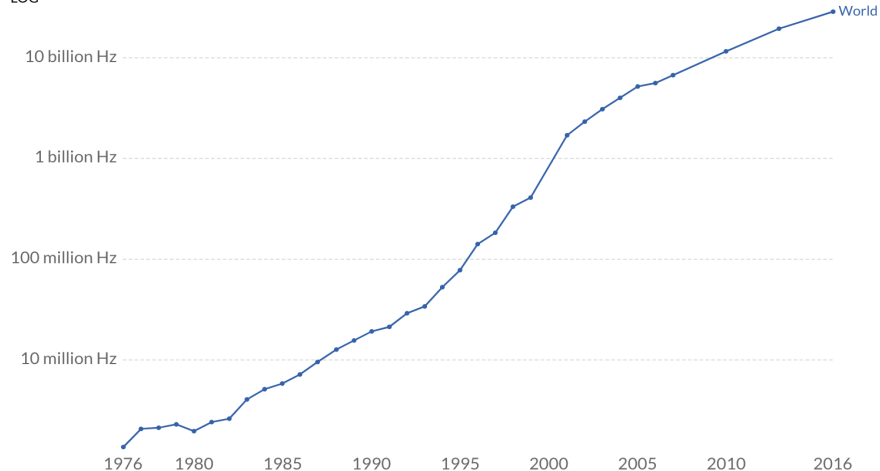
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Microprocessor clock speed

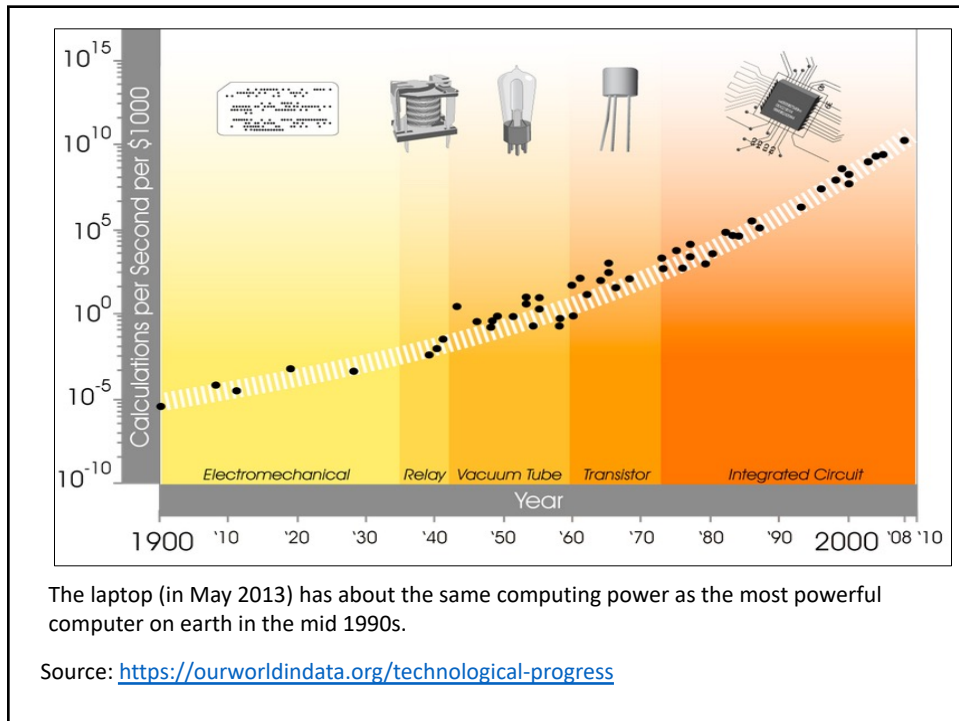
Microprocessor clock speed measures the number of pulses per second generated by an oscillator that sets the tempo for the processor. It is measured in hertz (pulses per second).

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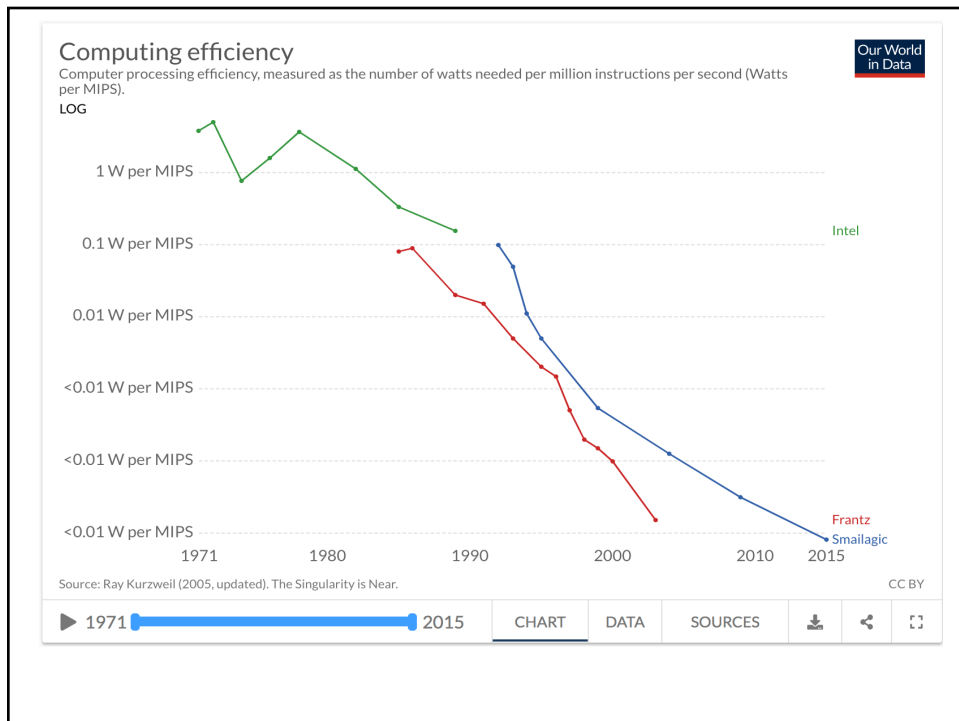


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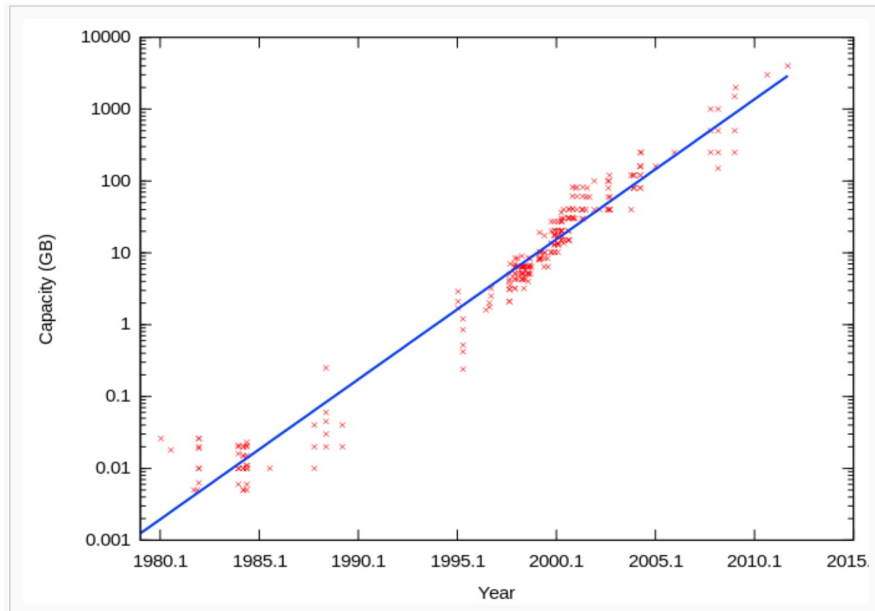
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References & Sources of Images

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Thank you!